2. Mapping the Field: Five Methodological Clusters

Reading across fifteen papers on the same broad problem, I relised that they are not all talking about the same thing even when they use the same words. "Generalisation failure" means something different to a genomics researcher checking for data leakage than it does to a philosopher distinguishing statistical extrapolation from practical robustness.

Grouping the fifteen sources into five clusters. Each cluster shares a common way of framing the problem, and understanding that framing also means understanding its blind spots.

Cluster 1 Pitfalls, Practice, and Reporting Standards	**Cluster 2 Dataset Shift and Distribution Mismatch**	**Cluster 3 Causal and Invariant Learning**	**Cluster 4 Generalisation Theory and Robustness Philosophy**	**Cluster 5 Clinical Translation and Illusory Validity**
Whalen et al. (2022)- Navigating the pitfalls of applying ML in genomics. Nature Reviews Genetics.	Wörheide et al. (2021)- Preventing dataset shift from breaking ML biomarkers. GigaScience.	Arjovsky et al. (2019)- Invariant Risk Minimization. arXiv.	Sullivan (2023)- Beyond generalisation: a theory of robustness in ML. Synthese.	Christodoulou et al. (2024)- Illusory generalisability of clinical prediction models. Science.
Collins et al. (2024)- TRIPOD+AI statement. BMJ.	Springer & Doering (2021)- Unseen-epitope TCR interaction prediction. Briefings in Bioinformatics.	Emerson et al. (2023)- Causal modelling for immune receptor diagnostics. Nature Machine Intelligence.	Watson & Floridi (2016)- [spans Clusters 3 and 4]	Collins et al. (2024)- [spans Clusters 1 and 5]
Riley et al. (2021)- Sample size for clinical prediction models. BMJ.	Ben-David et al. (2010)- A theory of learning from different domains. Machine Learning.	Watson & Floridi (2016)- Evolution and learning theory generalisation. PLOS Computational Biology.	Gretton et al. (2012)- A kernel two-sample test. JMLR.	Steyerberg et al. (2010)- Assessing prediction model performance. Epidemiology.
Vayena et al. (2018)- Machine learning in medicine: ethical challenges. PLOS Medicine.	Subbaswamy & Saria (2020)- Dataset shift, causality, and shift-stable models. Biostatistics.			

Cluster 1: Pitfalls, Practice, and Reporting Standards

The papers here share a particular belief: that most generalisation failures happen because researchers make avoidable technical mistakes.

Whalen et al. (2022), writing in *Nature Reviews Genetics*, give perhaps the most practically useful account of what those mistakes look like. They identify five specific ways that genomics ML studies go wrong: distributional differences between datasets, examples that are not truly independent of each other, hidden confounders, data leakage during preprocessing, and imbalanced class distributions. What makes this paper stand out is that they do not just describe the problems - they built interactive computational notebooks where we can see the performance inflation happen in real time on our own data. That kind of reproducibility is rare in review articles and genuinely useful.

Collins et al. (2024), the TRIPOD+AI statement in the *BMJ*, approach the same problem from a different direction. Rather than telling researchers what not to do, they specify what must be reported - a 27-item checklist built through a two-round Delphi process with 200 participants from across disciplines, institutions, and countries. The logic is that if studies are reported completely and transparently, the research community can identify problems and correct them. Riley et al. (2021) and Vayena et al. (2018) sit alongside these two, contributing the sample size calculations and ethical considerations that any well-designed clinical prediction study also needs to get right.

The problem with this cluster is that it treats generalisation failure as fundamentally a discipline problem. Follow the right steps, report everything properly, and the model will generalise. That is true as far as it goes. But it leaves open the possibility that some failures are not about sloppiness at all. They happen even when every box is ticked, because the gap between the training population and the deployment population is too large for any amount of methodological rigour to bridge. A checklist cannot fix a structural mismatch in the data-generating process.

2.2 Cluster 2: Dataset Shift and Distribution Mismatch

Goes a level deeper than Cluster 1. The problem isn't just researcher error - the world the model trained on is genuinely different from the world it gets deployed into.

Ben-David et al. (2010) is the theoretical anchor. The key result: transferability depends on (a) how different the two distributions are, and (b) how well any classifier could perform on both simultaneously. The implication is that in some cases the divergence is so large that no correction strategy works. That's a hard theoretical limit, not an engineering problem to solve. Worth flagging that the applied papers in this cluster tend to underweight this.

Wörheide et al. (2021) bring it into clinical biomarker work with the covariate shift vs. prior probability shift distinction. Important - these aren't just different names, they require different corrections. Conflating them would lead to applying the wrong fix.

Springer and Doering (2021) push further into zero-shot territory - predicting interactions for biological targets the model has never seen at all. Not just new population, genuinely unseen targets. That's a harder version of the problem.

Subbaswamy and Saria (2020) act as a bridge toward Cluster 3 - arguing the only robust solution is identifying causally stable mechanisms rather than endlessly re-weighting unstable statistical associations. That's a significant claim.

The tension noted between Ben-David et al.'s theoretical pessimism and the applied papers' relative optimism about correction strategies is real. Neither group has tested their positions head-to-head on the same dataset. Both are more confident than the combined evidence justifies.

2.3 Cluster 3: Causal and Invariant Learning

The most radical argument so far. The claim: standard ML learns whatever associations reduce training error. If age correlates with the outcome in training data, the model uses age - even if that correlation is spurious and won't hold elsewhere. The model has no mechanism for distinguishing genuine causal relationships from coincidental ones.

Arjovsky et al. (2019) - IRM. The logic: find features that predict the outcome equally well across multiple different training environments. Consistency across contexts is evidence of a genuine mechanism rather than a local statistical artifact.

Emerson et al. (2023) apply this to immune receptor diagnostics using causal graphs (DAGs) and controlled simulation. Grounded application rather than pure theory.

Watson and Floridi (2016) are the most ambitious here - using PAC learning theory to argue the generalisation problem in ML mirrors the generalisation problem in biological evolution. If that analogy holds, the problem isn't specific to ML methodology at all but reflects something fundamental about how any adaptive system transfers learning across environments. That's either a very deep insight or an overreach - would need to look more carefully at the PAC learning argument.

Practical limitation is significant and acknowledged: IRM needs multiple distinct training environments, which clinical data almost never provides. Causal graphs require substantial domain expertise upfront. The gap between the theory and what a clinical ML team can actually implement is wide. Exciting intellectually but not yet practically accessible.

2.4 Cluster 4: Generalisation Theory and Robustness Philosophy

This is the most conceptually challenging cluster, and possibly the most important one for understanding why the field keeps running into the same problems.

Sullivan (2023), writing in *Synthese*, makes a distinction that sounds subtle but has large practical consequences. Statistical generalisation - demonstrating that model works on a held-out sample from the same population - is not the same thing as practical robustness - demonstrating that model performs reliably when the population changes, the data collection process shifts, or the deployment environment evolves. These are two different properties, and they require two different kinds of evidence. The problem, Sullivan argues, is that clinical ML routinely provides evidence for the first while claiming to have demonstrated the second. Cross-validation on a single dataset tells something real, but it does not tell what happens when the model moves to a new hospital.

Gretton et al. (2012), in a more technical contribution from the *Journal of Machine Learning Research*, provide a rigorous statistical test - the Maximum Mean Discrepancy (MMD) kernel test - for detecting whether two datasets come from the same distribution. This might sound narrow, but it is actually important: before I can correct for shift, need a principled way to detect and quantify it. That is what this work provides.

Together, these papers question the conceptual foundations of how the field measures success, rather than just the technical execution.

2.5 Cluster 5: Clinical Translation and Illusory Validity

The papers look at what actually happens when clinical prediction models are deployed.

Christodoulou et al. (2024), in *Science*, provide the most direct evidence in the entire corpus. They took models trained to predict antipsychotic treatment outcomes in schizophrenia, tested them within the trials they were trained on, and then tested them on independent trial data. Within training, the models averaged 0.72 balanced accuracy - genuinely impressive. On independent data, they averaged 0.54, which is barely better than flipping a coin. The models had not learned anything that generalised. They had learned something specific to the statistical quirks of each trial.

Steyerberg et al. (2010) provide the classical framework for thinking about what good prediction model performance actually looks like - discrimination (can the model separate high-risk from low-risk patients?), calibration (are the predicted probabilities actually accurate?), and clinical utility (does using the model improve decisions?). These three criteria are the standard against which Christodoulou et al.'s failures are so striking. Collins et al. (2024) close the loop between Clusters 1 and 5: the TRIPOD+AI reporting requirements exist precisely to prevent the kind of invisible failure that Christodoulou et al. document.

The shared limitation of this cluster is that it is very good at showing that models fail and documenting the shape of that failure, but relatively quiet about what to do about it prospectively. Knowing a model failed after the fact is useful. Knowing before deployment whether it is likely to fail would be transformative. That prospective capacity is what is missing - research gap